



This sustainability chatbot is powered by Amazon Nova Pro, using data from NASA, IRENA, IEA, NSIDC and NOAA



Your Sustainability Chatbot - Powered by Amazon Bedrock with Nova Pro

Quick Questions:

Explain in Japanese the temperature trends

Explain to a toddler what are critical minerals and why they matter

What is the current CO2 level, and how does that impact me?

When will we see an Ice-free Summer? And what does that mean?

Hello! I'm your AI climate assistant. Ask me about climate data, renewable energy, or environmental trends.



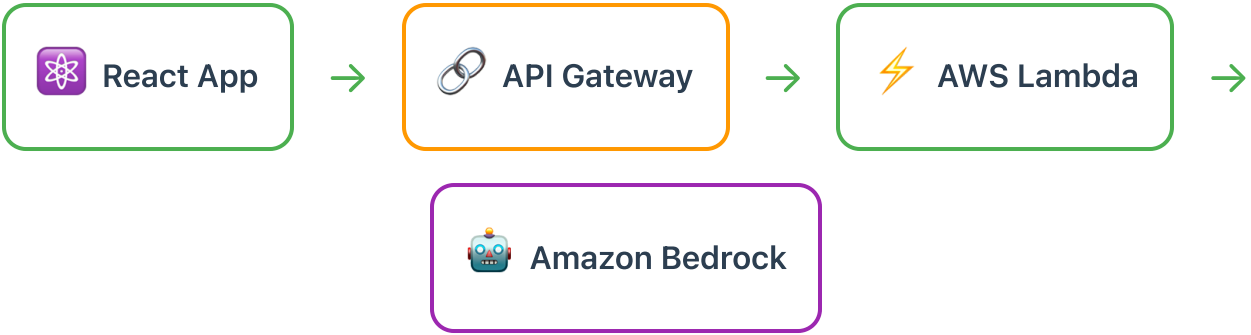
Sustainability Chatbot

Ask about climate data, renewable energy, or environmental trends...

Send



Technical Architecture Flow



AI Model

Amazon Nova Pro



Data Storage

Amazon S3



Security

AWS IAM

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

Technology & AI

Frontend: React.js, D3.js, Chart.js, Framer Motion

Backend: Python Flask, NumPy, Pandas

AI Integration: Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation

AI Chatbot: Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM

Deployment: AWS Amplify with automated CI/CD pipeline

Data Sources



Temperature Data:

NASA GISTEMP v4 (Goddard Institute)



Renewable Energy:

IRENA Global Energy Statistics



Critical Minerals:

IEA Critical Minerals Data Explorer



Sea Ice Data:

NSIDC Sea Ice Index



CO2 Data:

NOAA Mauna Loa Observatory

Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics

Challenge: "Visualizing a Sustainable Tomorrow"

Approach: Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data

Goal: Inspire climate action through compelling data storytelling

Features: Interactive visualizations, correlation analysis, predictive modeling, and accessibility-first design

Built with  by [@ygal](#) and his best friend Amazon Q 

Powered by  | NASA, IRENA, IEA, NSIDC & NOAA data